Sinusoidal Initialization: Time for a New Start

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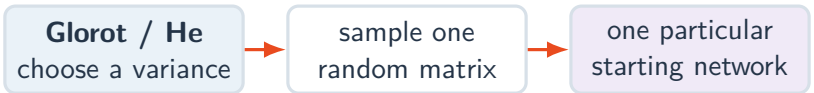
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A good scale is not a complete starting point



Controlled
signal scale

Left to chance
balance inside each neuron

The variance can be right while the starting point is not.

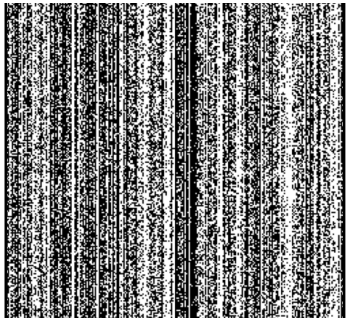
Some neurons start almost always ON — or OFF

One neuron across eight inputs



The neuron barely uses its nonlinearity.

We call this a **skewed neuron**.



Each column is a neuron. Dark or light columns reveal one-sided behaviour.

A network can begin with part of its capacity already underused.

One number reveals the imbalance

$$S_i = \sum_{j=1}^{n} W[i,j]$$

sum of the incoming weights

A zero-mean distribution does not force each row to sum to zero.

The theoretical link

Large $|S_i|$ makes a neuron more likely to prefer one side of its activation.

The random case

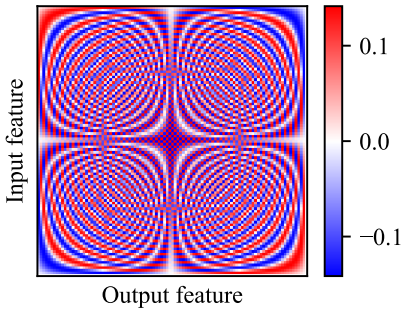
With standard zero-mean random weights, skewed neurons become likely as fan-in grows.

So the design target is simple: make every S_i equal to zero.

Build each row as a wave

$$W[i,j] = a\sin(k_i x_j + \phi_i)$$

$$k_i = 2\pi i, \quad x_j = j/n, \quad \phi_i = 2\pi i/m$$



- one frequency per neuron
- one phase per neuron
- amplitude a matches Glorot variance

Rows are different, but none is sampled at random.

Same scale. Deliberate structure. No seed dependence.

The wave cancels itself

$$\sum_{j=1}^{n} W[i,j] = a \sum_{j=1}^{n} \sin\left(2\pi i \frac{j}{n} + \phi_i\right) = 0$$

Exact balance
in every row

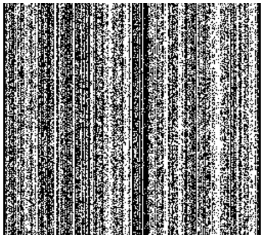
Different waves
across neurons

Correct variance
through amplitude

This is the whole construction: balance, diversity, and scale.

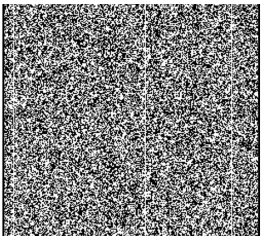
What changes inside the network?

He normal



One-sided responses
many columns stay dark or light

Sinusoidal



More balanced responses
neurons switch across inputs

Five very different tests

ResNet-50	MobileNetV3	EfficientNetV2	ViT-B16	BERT-mini
CIFAR-100	Tiny-ImageNet	Tiny-ImageNet	ImageNet-1K	WikiText

4 initializations
Default, Orthogonal, LSUV, Sinusoidal

→

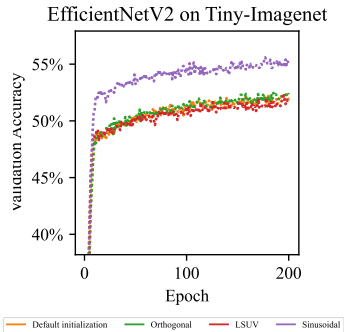
Train from scratch
same budget per comparison

→

Measure two outcomes
final accuracy and training AUC

The question is not whether one model improves, but whether the pattern transfers.

The main result: faster, then better



EfficientNetV2 on Tiny-ImageNet.

+20.9%

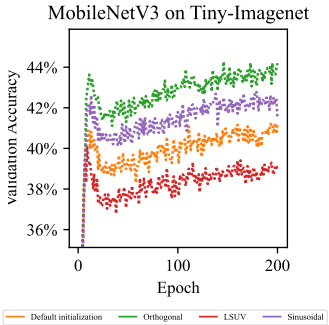
average training AUC vs. default

+4.9%

average final accuracy vs. default

The advantage appears early and often remains at the end.

One exception tells us something



MobileNetV3 on Tiny-ImageNet.

With HardSwish

Orthogonal is slightly better.

With ReLU

Sinusoidal leads again.

The activation function can change which initialization works best.

What should we remember?

1. **Variance is only part of initialization.**
2. A sinusoidal structure gives each neuron **exact row balance** and **distinct responses**.
3. Across CNNs, a ViT, and BERT, it usually learns **faster** and ends **better**.

Randomness is not a requirement for a good start.

Fernández-Hernández et al., *Sinusoidal Initialization, Time for a New Start*, NeurIPS 2025.

Questions?